YU-HAN HUANG

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PROFILE

I'm a senior undergraduate student in Electrical Engineering at NTU with research experience in speech and multimodal AI. Through my current work in speech and vision laboratories, I have gained exposure to questions related to model reasoning, multimodal understanding, and VLA models. My long-term goal is to pursue graduate studies and develop the ability to conduct independent research on Machine Learning.

EDUCATION

B.S. in Electrical Engineering, National Taiwan University (NTU), Taipei, Taiwan *2023–Present*

- avg GPA: 4.05/4.3 (equivalent to 3.93/4.0)
- Selected Coursework: Machine Learning, Deep Learning for Computer Vision, Deep Reinforcement Learning

RESEARCH EXPERIENCE

Visiting Scholar, NLP Lab, University of California, Los Angeles (UCLA) *Summer 2026*

- Conducting research on AI agents, with a focus on GUI agents

Undergraduate Research, Speech Processing and Machine Learning Lab, NTU *2025–Present*

- Supervisor: Prof. Hung-Yi Lee
- Exploring training-free approaches to improve audio model understanding and response reliability
- Surveying recent literature on multimodal language models, reasoning, and post-training methods

Undergraduate Research, Vision and Learning Lab, NTU *2025–Present*

- Supervisor: Prof. Yu-Chiang Frank Wang
- Studying VLA models for long horizon embodied task planning
- Contributing to ongoing research through experiment implementation and empirical evaluation

Research Intern, ASUS Open Cloud Infrastructure Software *2026–Present*

- Surveying recent research on AI agent memory, long-context modeling, and LLM-based agents

Research Intern, Institute of Information Science, Academia Sinica, Taipei *Summer 2024*

- Supervisor: Prof. Yennan Huang
- Developed automated pipelines for generating both attack and benign audit logs for intrusion detection research
- Supported research on attack scenario generation and security data simulation

PUBLICATIONS

- **Yu-Han Huang**, Chih-Kai Yang, Ke-Han Lu, Cheng An Yu, Hung-yi Lee.
  Enhancing How Large Audio-Language Models Hear: Identifying and Amplifying Acoustic Neurons in the Audio Encoder.
  IEEE SLT 2026, Under review.
- Lok-Lam Ieong, Chia-Chien Chen, Chih-Kai Yang, **Yu-Han Huang**, An-Yu Cheng, Hung-yi Lee.
  Nudging Hidden States: Training-Free Model Steering for Chain-of-Thought Reasoning in Large Audio-Language Models.
  Interspeech 2026.
- Kai-Wei Chang, Yi-Cheng Lin, Huang-Cheng Chou, Wenze Ren, **Yu-Han Huang**, Yun-Shao Tsai, Chien-Cheng Chen, Yu Tsao, Yuan-Fu Liao, Shrikanth Narayanan, James Glass, Hung-yi Lee.
  TaigiSpeech: A Low-Resource Real-World Speech Intent Dataset and Preliminary Results with Scalable Data Mining In-the-Wild.
  Interspeech 2026.
- Ke-Han Lu, Szu-Wei Fu, Chao-Han Huck Yang, Zhehuai Chen, Sung-Feng Huang, Chih-Kai Yang, Yi-Cheng

Lin, Chi-Yuan Hsiao, Wenze Ren, En-Pei Hu, **Yu-Han Huang**, An-Yu Cheng, Cheng-Han Chiang, Yu Tsao, Yu-Chiang Frank Wang, Hung-yi Lee.
How Auditory Knowledge in LLM Backbones Shapes Audio Language Models: A Holistic Evaluation.
IEEE SLT 2026, Under review.

- Kai-Po Chang, Chang-Hsun Wu, Yen-Siang Wu, **Yu-Han Huang**, Kai-Chen Yang, Yu-Chiang Frank Wang
Self-Correcting Imagination Planning for VLM Embodied Agents.
CoRL 2026, Under review.

OTHER EXPERIENCES

Machine Learning Teaching Assistant, EECS Department, NTU *Spring 2026*

- Instructor: Prof. Hung-Yi Lee

MakeNTU Hackathon – 1st Place, Enterprise Award, Taipei, Taiwan *2025*

- Project: **AI Driving Coach**
- Integrated WebSocket for real-time data streaming and Google Gemini API for driver behavior analysis
- Developed real-time voice feedback mechanisms and data visualization

National Taiwan Bionic Design Competition – 2nd Place *2022*

SKILLS

Python, PyTorch, Machine Learning, Speech Processing, Linux, Git, Docker, AI coding tools